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Studierichting: Informatica
Oefeningenreeks 2D nr. 10.7

$$\lim_{x\to -2\sqrt{5}} \frac{\sqrt{5}x^2+9x-2\sqrt{5}}{2x^2+3x\sqrt{5}-10} = \frac{0}{0}$$

We noemen de vergelijking in de teller en noemer respectievelijk $t(x)$ en $n(x)$

$x_{1,2}$ van $t(x)$:

$D = 81 - 4 \cdot \sqrt{5} \cdot \left(-2\sqrt{5}\right)$

$= 81 + 40 = 121$

$\sqrt{D} = 11$

$x_{1,2} = \dfrac{-9 \pm 11}{2\sqrt{5}} \Rightarrow \dfrac{1}{\sqrt{5}}$ en $-2\sqrt{5}$

$t(x) = \sqrt{5}(x-x_1)(x-x_2)$

$= \sqrt{5}\left(x - \dfrac{1}{\sqrt{5}}\right)\left(x+2\sqrt{5}\right)$

$= \left(\sqrt{5}x-1\right)\left(x+2\sqrt{5}\right)$

$x_{1,2}$ van $n(x)$:

$D = \left(3\sqrt{5}\right)^2 - 4 \cdot 2 \cdot (-10)$

$= 9 \cdot 5 + 80$

$= 45 + 80 = 125$

$\sqrt{D} = 5\sqrt{5}$

$x_{1,2} = \dfrac{-3\sqrt{5} \pm 5\sqrt{5}}{4} \Rightarrow \dfrac{\sqrt{5}}{2}$ en $-2\sqrt{5}$

$n(x) = 2(x-x_1)(x-x_2)$

$= 2\left(x - \dfrac{\sqrt{5}}{2}\right)\left(x+2\sqrt{5}\right)$

$= \left(2x-\sqrt{5}\right)\left(x+2\sqrt{5}\right)$

$$\lim_{x\to -2\sqrt{5}} \frac{t(x)}{n(x)}$$

$$= \lim_{x\to -2\sqrt{5}} \frac{\left(\sqrt{5}x-1\right)\cancel{\left(x+2\sqrt{5}\right)}}{\left(2x-\sqrt{5}\right)\cancel{\left(x+2\sqrt{5}\right)}}$$

$$
\begin{aligned}
&= \lim_{x\to -2\sqrt{5}} \frac{\sqrt{5}x - 1}{2x - \sqrt{5}} \\
&= \frac{\left(\sqrt{5}(-2\sqrt{5}) - 1\right)}{\left(2(-2\sqrt{5}) - \sqrt{5}\right)} \\
&= \frac{-2 \cdot 5 - 1}{-4\sqrt{5} - \sqrt{5}} \\
&= \frac{-11}{-5\sqrt{5}} \\
&= \frac{11\sqrt{5}}{25}
\end{aligned}
$$

## References